

Calendar — Tricks, Shortcuts & Question Patterns

Anchor-Day Method, Month Codes & What Repeats in TCS / CAT

Aptitude Grind

1 The Anchor-Day Method (Fastest for Placement Tests)

Instead of counting odd days from year 1, work relative to a **known reference date** that is close to your target date. Most questions either give you a reference day directly, or you can use a memorized anchor.

Useful Anchors to Memorize

- 1 January, 1 AD $\rightarrow$ Monday
- 1 January, 1600 (or 2000) $\rightarrow$ Saturday
- 1 January, 2000 $\rightarrow$ Saturday (most practical anchor for modern-date questions)

Anchor-Day Procedure

1. Find the number of complete years between the anchor and target year.
2. Odd days contributed = (ordinary years $\times$ 1) + (leap years $\times$ 2), reduced mod 7.
3. Add odd days for the months elapsed in the target year, then add the date number.
4. Add this to the anchor's day-of-week position (mod 7) and read off the result.

Example — Using an Anchor

If 1 Jan 2000 was a Saturday, what day was 1 Jan 2010?

- Years 2000–2009 (10 years, but we count odd days contributed *up to* 1 Jan 2010, i.e. years 2000–2009 inclusive as full years elapsed)
- Leap years in that span: 2000, 2004, 2008 $\rightarrow$ 3 leap years; ordinary years: 7
- Odd days = $(3 \times 2) + (7 \times 1) = 13 \equiv 6 \pmod{7}$
- Saturday +6 odd days $\rightarrow$ **Friday**

2 Month Codes Method (a.k.a. Doomsday-style Shortcut)

A faster alternative used heavily in coaching materials: assign each month a fixed **code** representing its odd-day contribution from 1 January of a non-leap year up to the 1st of that month.

Month Codes — Non-Leap Year

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Code	0	3	3	6	1	4	6	2	5	0	3	5

Leap Year Adjustment

For a **leap year**, subtract 1 from the codes of **January and February only** (since the extra day, 29 Feb, hasn't occurred yet when computing odd days up to those months). From March onward in a leap year, use the codes as-is since the leap day has already been "absorbed."

Putting It Together

$$\text{Day code} = (\text{Date} + \text{Month code} + \text{Year's odd days from anchor century}) \bmod 7$$

Match the result against the Odd-Days $\rightarrow$ Day table from `01-fundamentals.tex`.

When to Use Which Method

The **odd-days-from-scratch** method (Notes 1) is more intuitive and safer under exam pressure because you derive everything instead of recalling a code table. The **month-code** method is faster once memorized, but a single misremembered code silently gives a wrong answer. For placement tests with negative marking, prefer the method you can execute without doubt.

3 Same Calendar Repeats

Rule

Two years have the **same calendar** if:

1. Both are ordinary years or both are leap years, **and**
2. The total number of odd days between them (summed year by year) is a multiple of 7.

Example

Find the next year with the same calendar as 2007 (an ordinary year).

- 2007 $\rightarrow$ 2008: ordinary year, 1 odd day. Running total: 1
- 2008 $\rightarrow$ 2009: 2008 is leap, 2 odd days. Running total: 3
- 2009 $\rightarrow$ 2010: ordinary, 1 odd day. Running total: 4
- 2010 $\rightarrow$ 2011: ordinary, 1 odd day. Running total: 5
- 2011 $\rightarrow$ 2012: ordinary, 1 odd day. Running total: 6

- 2012 → 2013: 2012 is leap, 2 odd days. Running total: $8 \equiv 1$
- Continue... total odd days reach a multiple of 7 at **2018** (11 years later)

This is a classic recurring question type: *"Calendar for 2007 will be the same as calendar for which year?"* → Answer: **2018**.

Speed Rule of Thumb

For an ordinary year to repeat, you typically need **6, 11, or 17 years** later (rarely other gaps), and the target year must also be ordinary. For a leap year, it usually repeats after **28 years** (or sometimes 12/40 depending on century boundaries). Don't blindly memorize this — verify with the running-odd-day-total method above, since century boundaries (1800, 1900, 2100) break the usual pattern.

4 Common Question Types in TCS NQT, CAT & Similar Exams

Based on recurring patterns across placement and competitive exams, Calendar questions almost always fall into one of these categories:

Category 1: Find the Day for a Given Date

"What day of the week was 15 August 1947?" — direct application of the odd-days method. Very common as a standalone question in TCS NQT and SSC/Bank exams.

Category 2: Day N Days After/Before a Given Day

"Today is Monday. What day will it be after 61 days?" — take $N \bmod 7$ and count forward (or backward) from the given day. **No leap year logic needed** unless the range spans a different-length month setup (irrelevant here since we're just counting days, not months).

Category 3: Same Calendar / Repeating Year

"The calendar for 2005 is the same as the calendar for which year?" — use the running-odd-day-total method above.

Category 4: Given One Date's Day — Find Another Date's Day (Different Year)

"If 8 Dec 2007 was Saturday, what day was 8 Dec 2006?" — find odd days *between* the two dates (same month/date, different year) and shift accordingly. Because month and date match, you only need to account for the number of odd days contributed by the intervening year(s) — no month calculation required. This is the fastest sub-type once you recognize it.

Category 5: Century / Leap-Year Reasoning (CAT-style — trickier)

"Can the last day of a century ever be a Tuesday?" — requires knowing that centuries (100, 200, 300, 400 years) only ever contribute 5, 3, 1, or 0 odd days, so the last day of *any* century can only be **Sunday, Monday, Wednesday, or Friday** — never Tuesday, Thursday, or Saturday. CAT and other reasoning-heavy exams like to test this kind of derived property rather than a direct date lookup.

Category 6: Reverse Problems — Find the Year or Date

"In which year after 2000 will 1 January fall on a Sunday, given some constraint?" — work backward from the odd-day requirement. Less common but appears in trickier CAT-adjacent sets.

5 Exam Strategy Summary

- Identify the category first — it tells you which shortcut applies (don't default to full odd-day counting from year 1 if a same-date/different-year shortcut applies).
- For "N days after/before" questions, just do $N \bmod 7$ — don't overthink with leap years.
- For "same calendar" questions, track the running odd-day total year by year; stop at the first multiple of 7.
- Always double-check century years (1800, 1900, 2100, ...) for the leap year exception — these are the most common trap in tricky questions.
- Practice the odd-days method until it's automatic — speed comes from not having to think about *why* each step works, only executing it.